

PROBABILITY THEORY AND RANDOM PROCESSES (MA225)

LECTURE SLIDES

Lecture 17

Modes of Convergence

Let $\{X_n\}$ be a sequence of random variables defined on a probability space $(\mathcal{S}, \mathcal{F}, P)$. Let X be a random variable defined on the same probability space $(\mathcal{S}, \mathcal{F}, P)$.

Def: (Almost sure convergence) We say that X_n converges almost surely or with probability 1 to a random variable X if

$$P(\omega : X_n(\omega) \rightarrow X(\omega)) = 1.$$

Example 1: Let $\mathcal{S} = [0, 1]$, $\mathcal{F} = \mathcal{B}([0, 1])$ and P be the uniform measure. Define $X_n = 1_{[0, \frac{1}{n}]}$. Then X_n converges almost surely (w. p. 1) to the zero random variable.

Theorem: Let $\{X_n\}$ and $\{Y_n\}$ be two sequences of random variables defined on a probability space $(\mathcal{S}, \mathcal{F}, P)$. Suppose $X_n \rightarrow X$ w. p. 1 and $Y_n \rightarrow Y$ w. p. 1. Then

- $X_n + Y_n \rightarrow X + Y$ w. p. 1.
- $X_n Y_n \rightarrow XY$ w. p. 1.
- $f(X_n) \rightarrow f(X)$ w. p. 1, for any f continuous.

Def: (Convergence in r^{th} mean) We say that X_n converges in r^{th} mean to a random variable X if

$$E|X_n - X|^r \rightarrow 0.$$

Example 2: Let $\mathcal{S} = [0, 1]$, $\mathcal{F} = \mathcal{B}([0, 1])$ and P be the uniform measure. Define $X_n = 1_{[0, \frac{1}{n}]}$. Then X_n converges in r^{th} mean to the zero random variable.

Theorem: Let $\{X_n\}$ and $\{Y_n\}$ be two sequences of random variables defined on a probability space $(\mathcal{S}, \mathcal{F}, P)$.

- If $X_n \rightarrow X$ in r^{th} mean and $Y_n \rightarrow Y$ in r^{th} mean, then $X_n + Y_n \rightarrow X + Y$ in r^{th} mean.
- If $X_n \rightarrow X$ in r^{th} mean then $f(X_n) \rightarrow f(X)$ in r^{th} mean, for any f bounded continuous.

Def: (Convergence in probability) We say that X_n converges in probability to a random variable X if for any $\epsilon > 0$,

$$P(|X_n - X| > \epsilon) \rightarrow 0.$$

Example 3: Let $\mathcal{S} = [0, 1]$, $\mathcal{F} = \mathcal{B}([0, 1])$ and P be the uniform measure. Define $X_n = n1_{[0, \frac{1}{n}]}$. Then X_n converges in probability to the zero random variable.

Theorem: Let $\{X_n\}$ and $\{Y_n\}$ be two sequences of random variables defined on a probability space $(\mathcal{S}, \mathcal{F}, P)$. Suppose $X_n \rightarrow X$ in probability and $Y_n \rightarrow Y$ in probability. Then

- $X_n + Y_n \rightarrow X + Y$ in probability.
- $X_n Y_n \rightarrow XY$ in probability.
- $f(X_n) \rightarrow f(X)$ in probability, for any f continuous.

Def: (Convergence in distribution) We say that X_n converges in distribution to a random variable X if

$$F_n(x) \rightarrow F(x)$$

for all x where F is continuous and where F_n s are the distribution functions of X_n s and F is the distribution function of X .

Remark: Unlike the first three modes of convergence, here X_n s can be defined on different probability spaces. We are only interested in the distribution functions. This flexibility makes this mode of convergence very useful.

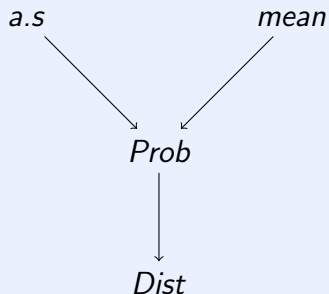
Example 4: Suppose X_n s are random variables such that $P(X_n = 1/n) = 1$. Then X_n converges in distribution to the zero random variable.

Theorem: Let $\{X_n\}$ and $\{Y_n\}$ be two sequences of random variables defined on a probability space $(\mathcal{S}, \mathcal{F}, P)$. Suppose $X_n \rightarrow X$ in distribution and $Y_n \rightarrow c$ in probability for some constant c . Then

- $X_n + Y_n \rightarrow X + c$ in distribution.
- $X_n Y_n \rightarrow cX$ in distribution.
- $f(X_n) \rightarrow f(X)$ in distribution, for any f continuous.

Important: If X_n converges to X in distribution and Y_n converges to Y in distribution then $X_n + Y_n$ may not converge to $X + Y$ in distribution. Same for product.

Relation between Modes of Convergence



Theorem: Suppose $\{X_n\}$ is a sequence of RVs defined on a single probability space and X_n converges in distribution to some constant c , then X_n also converges in probability to c .

Theorem: Let $\{X_n\}$ be a sequence of random variables with moment generating functions $M_n(t)$. Let X be a random variable with moment generating function $M(t)$. If $M_n(t) \rightarrow M(t)$ for all t in an open interval containing zero, then X_n converges to X in distribution.

Example 5: Let $X_n \sim \text{Bin}(n, p_n)$, where $p_n \rightarrow 0$ and $np_n = \lambda(> 0)$. Let $X \sim \text{Poi}(\lambda)$. Then X_n converges to X in distribution.

Theorem: Let $\{X_n\}$ be a sequence of DRV's with PMF $f_n(\cdot)$. Let X be a DRV with PMF $f(\cdot)$. If $f_n(x) \rightarrow f(x)$ for all x , then X_n converges to X in distribution.

Example 6: Prove the claim of the previous example using the above Theorem.

Theorem: Let $\{X_n\}$ be a sequence of CRV's with PDF $f_n(\cdot)$. Let X be a CRV with PDF $f(\cdot)$. If $f_n(x) \rightarrow f(x)$ for all x , then X_n converges to X in distribution.

Example 7: Let $X_n \sim U(0, 1 + 1/n)$ and $X \sim U(0, 1)$. Then X_n converges to X in distribution.